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Qualification Pack



Junior Engineer - Signal (Metro Rail)

QP Code: DMR/LSC/Q0201

Version: 1.0

NSQF Level: 4.5

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DMR/LSC/Q0201: Junior Engineer - Signal (Metro Rail)

Brief Job Description

A Junior Engineer - Signal (Metro Rail) oversees metro rail signalling systems' installation, maintenance, and troubleshooting to ensure safe and efficient train operations. The individual's responsibilities include overseeing point machines, track circuits, interlocking systems, and Platform Screen Doors (PSDs) and coordinating with telecom and power supply teams. It also involves monitoring Automatic Train Protection (ATP) and Automatic Train Supervision (ATS) systems, ensuring compliance with safety protocols, and conducting regular maintenance and emergency response drills.

Personal Attributes

The individual should be detail-oriented, safety-conscious, and possess strong analytical and problem-solving skills. Excellent communication and teamwork abilities are essential for coordinating with cross-functional teams and ensuring operational efficiency.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [DMR/LSC/N9901: Implement metro operations training and safety procedures](#)
2. [DMR/LSC/N0201: Oversee the operation and maintenance of metro signalling and train control systems](#)
3. [DMR/LSC/N0202: Oversee the operation and maintenance of signalling and safety systems for rail operations](#)
4. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Logistics
Sub-Sector	Transport (Metro Rail)
Occupation	Signalling & Telecom
Country	India
NSQF Level	4.5
Credits	17

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Aligned to NCO/ISCO/ISIC Code	NCO-2015/2153.9900
Minimum Educational Qualification & Experience	Diploma (3 years Engineering Diploma / equivalent in Electronics, Computer Engg., Information Technology, Electronics & Communications, Electronics & Communications Industry Integrated, Electrical & Electronics, Electronics / Microprocessor, Electronics & Telecommunications, Instrumentation Technology, Electronics and Instrumentation Engg. or in relevant field)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	18/02/2028
NSQC Approval Date	18/02/2025
Version	1.0
Reference code on NQR	QG-4.5-TW-03512-2025-V1-DMRCA
NQR Version	1.0

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DMR/LSC/N9901: Implement metro operations training and safety procedures

Description

This unit focuses on implementing structured training and safety measures for metro operations, ensuring technical knowledge, system readiness, and emergency preparedness.

Scope

The scope covers the following :

- Training Module Overview
- Safety Overview
- Store Management and Procurement
- Overview of Telecom, Signaling, and Track Systems
- Overview of Traction, SCADA, and E&M Systems
- OCC and Rolling Stock Overview
- Operations and Customer Care
- Practical Visits
- First Aid Training
- Fire Fighting Systems
- Security in Metro Systems
- Metro Rail General Rules 2020
- Safety Circulars and Contingency Plans

Elements and Performance Criteria

Training Module Overview

To be competent, the user/individual on the job must be able to:

- PC1.** follow schedules for weekly activities and guidelines on punctuality, discipline, and conduct.
- PC2.** conduct identity verification and maintain attendance records
- PC3.** ensure adherence to training timelines and program deliverables

Safety Overview

To be competent, the user/individual on the job must be able to:

- PC4.** identify the role of the safety department and follow measures for personal, passenger, and system safety
- PC5.** use appropriate Personal Protective Equipment (PPE) in all designated areas
- PC6.** follow the hazard identification and risk assessment processes

Store Management and Procurement

To be competent, the user/individual on the job must be able to:

- PC7.** perform store management activities, including stock handling, documentation, and interdepartmental coordination for procurement.
- PC8.** monitor inventory levels and ensure timely requisitions for required materials
- PC9.** maintain records of procurement processes and vendor communication

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Overview of Telecom, Signaling, and Track Systems

To be competent, the user/individual on the job must be able to:

- PC10.** identify telecom components (AFC, CCTV, PIDS, radio systems, clocks) and perform basic functional checks
- PC11.** identify signaling systems (CATC, CBTC) and observe/report signal functioning during visits.
- PC12.** identify P-Way components (rails, sleepers, crossings) and report track conditions through basic patrolling tasks
- PC13.** report malfunctions in telecom, signaling, or track systems using the correct reporting protocols

Overview of Traction, SCADA, and E&M Systems

To be competent, the user/individual on the job must be able to:

- PC14.** identify traction power and SCADA components and interpret sectioning diagrams
- PC15.** identify E&M systems (RSS, ASS, DG sets, UPS) and perform basic checks to understand their functionality
- PC16.** report unusual system performance and assist in routine inspections

OCC and Rolling Stock Overview

To be competent, the user/individual on the job must be able to:

- PC17.** familiarize with OCC systems, tools, and the role of OCC controllers
- PC18.** identify rolling stock components and their maintenance procedures
- PC19.** report rolling stock irregularities observed during operations

Operations and Customer Care

To be competent, the user/individual on the job must be able to:

- PC20.** familiarize with station operations, revenue management, and crew control functions
- PC21.** identify public complaint channels, possible causes, and resolve complaints using proper processes
- PC22.** assist in managing passenger inquiries and ensuring customer satisfaction

Practical Visits

To be competent, the user/individual on the job must be able to:

- PC23.** conduct visits to OCC, stations, depots, and tunnels to observe layouts, facilities, and operations
- PC24.** document key observations during practical visits for knowledge reinforcement
- PC25.** identify operational challenges during visits and suggest improvement areas

First Aid Training

To be competent, the user/individual on the job must be able to:

- PC26.** identify the role and scope of first aid, including key human body functions (e.g., circulatory, respiratory systems)
- PC27.** follow appropriate techniques for treating bleeding, fractures, burns, shock, and unconsciousness
- PC28.** perform artificial respiration, CPR, and safe transportation of injured persons using stretchers
- PC29.** participate in simulated first aid scenarios to enhance preparedness

Fire Fighting Systems

To be competent, the user/individual on the job must be able to:

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- PC30.** identify elements of combustion, classes of fire, and appropriate extinguishing methods
- PC31.** operate active/passive fire protection systems, including alarms, fire extinguishers, and fire hydrants
- PC32.** conduct practical drills and identify fire suppression equipment at metro stations
- PC33.** respond to fire incidents promptly, ensuring passenger and system safety

Security in Metro Systems

To be competent, the user/individual on the job must be able to:

- PC34.** identify security threats, red alert situations, and follow protocols for disaster management
- PC35.** handle theft cases, manage abandoned persons, and lodge FIRs as per protocol
- PC36.** coordinate with security personnel to mitigate security breaches effectively

Metro Rail General Rules 2020

To be competent, the user/individual on the job must be able to:

- PC37.** interpret key provisions under Metro Rail General Rules 2020, including signals, train speed limits, and safety protocols
- PC38.** follow operational procedures for maintenance, station control duties, and driverless train systems
- PC39.** ensure compliance with rules for safe and efficient train operations

Safety Circulars and Contingency Plans

To be competent, the user/individual on the job must be able to:

- PC40.** implement procedures for the imposition of TSR, axle counter reset, Train ID management, and Computer Aided Train Supervision (CATS) operation
- PC41.** follow guidelines for engineering possession and maintenance activities during non-revenue hours, including hot work, solar system testing/repair, and role-play exercises
- PC42.** ensure compliance with procedures for stopping trains at platforms and during operational disruptions, such as earthquakes, high winds, and floods
- PC43.** manage degraded mode operations, including train movement during interlocking failures
- PC44.** oversee the issue, revision, updating, and deletion of SOPs, SWO/DWO, and timetable trials for new sections
- PC45.** facilitate the initiation of revenue train services, including pilot train movement, ATO/UTO/ATB mode operations, and AFTC functionality checks
- PC46.** execute procedures for negotiating neutral sections, managing pantograph failures, and monitoring Panto-OHE interface cameras
- PC47.** oversee train stabling, depot entry/exit, train movement, and fitness checks
- PC48.** conduct emergency evacuation using UG facilities, mid-tunnel shafts, and walkways for detainment of passengers due to technical failure, explosion, or fire
- PC49.** operate and manage non-ATP vehicles such as RGM and CMV/CTMC and address incidents like rail fractures, weld failures, rollback scenarios, radio communication issues and resetting of passenger alarms in train
- PC50.** coordinate vehicle movement on link lines and rescue immobilized trains
- PC51.** respond to fire/smoke incidents, including UG train fires, ensuring passenger and staff safety
- PC52.** operate ETS/BLS plungers, remove foreign objects like kites from OHE, and follow OHE safety guidelines

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of punctuality, discipline, and structured training
- KU2.** the procedures for trainee verification and attendance record maintenance
- KU3.** the role of time management in achieving training milestones
- KU4.** the role of the safety department and station/depot safety protocols
- KU5.** the use and importance of PPE
- KU6.** the basics of hazard identification, risk management, and accident reporting
- KU7.** the basics of stores management (inventory control, documentation)
- KU8.** the procurement processes and coordination across departments
- KU9.** the procedures for vendor management and maintaining procurement transparency
- KU10.** the telecom components, including AFC, CCTV, clocks, and communication systems
- KU11.** the basics of signaling systems (CATC, CBTC) and reporting unusual signal observations
- KU12.** the key P-Way components, track patrolling, and reporting track conditions
- KU13.** the common malfunctions in telecom, signaling, and track systems
- KU14.** the traction power systems and sectioning diagrams
- KU15.** the importance of SCADA and its functionalities
- KU16.** the role of E&M systems (DG sets, UPS) and their use during emergencies
- KU17.** the key preventive maintenance checks for E&M systems
- KU18.** the OCC systems, tools, and controller responsibilities
- KU19.** the rolling stock components, features, and basic maintenance
- KU20.** the reporting and addressing of common rolling stock issues
- KU21.** the station operations, revenue management, and crew coordination
- KU22.** the causes of public complaints and grievance resolution processes
- KU23.** the techniques for effective customer service and passenger engagement
- KU24.** the observations and learning outcomes from OCC, station, depot, and tunnel visits
- KU25.** the identifying real-world challenges during visits and linking theory to practice
- KU26.** the basic principles of first aid and key human body functions
- KU27.** the management of bleeding, fractures, burns, and unconsciousness
- KU28.** the artificial respiration, CPR, and safe patient transportation
- KU29.** the importance of first aid preparedness during emergencies
- KU30.** the elements of combustion, fire classifications, and extinguishing methods
- KU31.** the principles of passive/active fire protection systems
- KU32.** the practical usage of fire extinguishers, hydrants, and drills
- KU33.** the emergency response procedures during fire incidents
- KU34.** the security protocols, red alert situations, and theft management processes
- KU35.** the protocols for managing abandoned persons and disaster situations
- KU36.** the role of security personnel in ensuring passenger and asset safety
- KU37.** the key terminologies, signal usage, and operational protocols

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- KU38.** the rules for maintenance works, train operations, and driverless train systems
- KU39.** the role of regulatory compliance in metro rail safety
- KU40.** the safety circulars and contingency plans for fire/security incidents
- KU41.** the importance of proactive contingency planning for emergency scenarios.
- KU42.** the importance role-playing and drills to improve safety and response performance
- KU43.** the procedures for TSR imposition, axle counter reset, Train ID management, and CATS operations
- KU44.** the engineering possession protocols for maintenance tasks during non-revenue hours, including solar system testing and hot work
- KU45.** the guidelines for safely halting train operations during emergencies, natural disasters, or technical failures
- KU46.** the steps for managing degraded mode operations and interlocking failures
- KU47.** the SOP and SWO/DWO lifecycle management, including revisions and timetable trials for new sections
- KU48.** the ATO, UTO, and ATB mode operations, emphasizing operational readiness and AFTC checks
- KU49.** the procedures for neutral section negotiation, pantograph damage management, and monitoring Panto-OHE interfaces using cameras
- KU50.** the best practices for train stabling, depot operations, and train fitness verification
- KU51.** the emergency evacuation protocols, focusing on passenger safety in UG and mid-tunnel situations
- KU52.** the guidelines for operating non-ATP vehicles, handling rail fractures, weld failures, and resetting alarms during rollbacks
- KU53.** the vehicle operations on link lines and immobilized train rescue procedures
- KU54.** the protocols for fire/smoke incidents, particularly UG fires, and ensuring safety during such incidents
- KU55.** the operation of ETS/BLS plungers and removal of foreign objects from OHE systems, along with OHE safety practices

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** convey clear and concise information to teams, junior engineers, and customers
- GS2.** effectively prioritize tasks and adhere to training schedules and operational timelines
- GS3.** identify operational challenges and propose practical solutions during training or real-time operations
- GS4.** assess system performance, identify malfunctions, and report anomalies accurately
- GS5.** work cohesively with peers, instructors, and junior engineers to meet training objectives
- GS6.** address passenger queries and manage grievances professionally
- GS7.** proactively identify risks, follow safety protocols, and contribute to hazard-free operations
- GS8.** adapt to diverse training environments, system layouts, and evolving operational scenarios
- GS9.** accurately maintain logs, records, and reports for training and practical activities

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- GS10.** monitor systems, equipment, and processes, identifying areas requiring intervention or improvement

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Training Module Overview</i>	2	2	-	1
PC1. follow schedules for weekly activities and guidelines on punctuality, discipline, and conduct.	-	-	-	-
PC2. conduct identity verification and maintain attendance records	-	-	-	-
PC3. ensure adherence to training timelines and program deliverables	-	-	-	-
<i>Safety Overview</i>	2	2	-	1
PC4. identify the role of the safety department and follow measures for personal, passenger, and system safety	-	-	-	-
PC5. use appropriate Personal Protective Equipment (PPE) in all designated areas	-	-	-	-
PC6. follow the hazard identification and risk assessment processes	-	-	-	-
<i>Store Management and Procurement</i>	2	2	-	1
PC7. perform store management activities, including stock handling, documentation, and interdepartmental coordination for procurement.	-	-	-	-
PC8. monitor inventory levels and ensure timely requisitions for required materials	-	-	-	-
PC9. maintain records of procurement processes and vendor communication	-	-	-	-
<i>Overview of Telecom, Signaling, and Track Systems</i>	2	2	-	1
PC10. identify telecom components (AFC, CCTV, PIDS, radio systems, clocks) and perform basic functional checks	-	-	-	-
PC11. identify signaling systems (CATC, CBTC) and observe/report signal functioning during visits.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. identify P-Way components (rails, sleepers, crossings) and report track conditions through basic patrolling tasks	-	-	-	-
PC13. report malfunctions in telecom, signaling, or track systems using the correct reporting protocols	-	-	-	-
<i>Overview of Traction, SCADA, and E&M Systems</i>	2	2	-	1
PC14. identify traction power and SCADA components and interpret sectioning diagrams	-	-	-	-
PC15. identify E&M systems (RSS, ASS, DG sets, UPS) and perform basic checks to understand their functionality	-	-	-	-
PC16. report unusual system performance and assist in routine inspections	-	-	-	-
<i>OCC and Rolling Stock Overview</i>	2	2	-	1
PC17. familiarize with OCC systems, tools, and the role of OCC controllers	-	-	-	-
PC18. identify rolling stock components and their maintenance procedures	-	-	-	-
PC19. report rolling stock irregularities observed during operations	-	-	-	-
<i>Operations and Customer Care</i>	2	2	-	1
PC20. familiarize with station operations, revenue management, and crew control functions	-	-	-	-
PC21. identify public complaint channels, possible causes, and resolve complaints using proper processes	-	-	-	-
PC22. assist in managing passenger inquiries and ensuring customer satisfaction	-	-	-	-
<i>Practical Visits</i>	1	1	-	0.5
PC23. conduct visits to OCC, stations, depots, and tunnels to observe layouts, facilities, and operations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. document key observations during practical visits for knowledge reinforcement	-	-	-	-
PC25. identify operational challenges during visits and suggest improvement areas	-	-	-	-
<i>First Aid Training</i>	1	1	-	0.5
PC26. identify the role and scope of first aid, including key human body functions (e.g., circulatory, respiratory systems)	-	-	-	-
PC27. follow appropriate techniques for treating bleeding, fractures, burns, shock, and unconsciousness	-	-	-	-
PC28. perform artificial respiration, CPR, and safe transportation of injured persons using stretchers	-	-	-	-
PC29. participate in simulated first aid scenarios to enhance preparedness	-	-	-	-
<i>Fire Fighting Systems</i>	1	1	-	0.5
PC30. identify elements of combustion, classes of fire, and appropriate extinguishing methods	-	-	-	-
PC31. operate active/passive fire protection systems, including alarms, fire extinguishers, and fire hydrants	-	-	-	-
PC32. conduct practical drills and identify fire suppression equipment at metro stations	-	-	-	-
PC33. respond to fire incidents promptly, ensuring passenger and system safety	-	-	-	-
<i>Security in Metro Systems</i>	1	1	-	0.5
PC34. identify security threats, red alert situations, and follow protocols for disaster management	-	-	-	-
PC35. handle theft cases, manage abandoned persons, and lodge FIRs as per protocol	-	-	-	-
PC36. coordinate with security personnel to mitigate security breaches effectively	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Metro Rail General Rules 2020</i>	1	1	-	0.5
PC37. interpret key provisions under Metro Rail General Rules 2020, including signals, train speed limits, and safety protocols	-	-	-	-
PC38. follow operational procedures for maintenance, station control duties, and driverless train systems	-	-	-	-
PC39. ensure compliance with rules for safe and efficient train operations	-	-	-	-
<i>Safety Circulars and Contingency Plans</i>	1	1	-	0.5
PC40. implement procedures for the imposition of TSR, axle counter reset, Train ID management, and Computer Aided Train Supervision (CATS) operation	-	-	-	-
PC41. follow guidelines for engineering possession and maintenance activities during non-revenue hours, including hot work, solar system testing/repair, and role-play exercises	-	-	-	-
PC42. ensure compliance with procedures for stopping trains at platforms and during operational disruptions, such as earthquakes, high winds, and floods	-	-	-	-
PC43. manage degraded mode operations, including train movement during interlocking failures	-	-	-	-
PC44. oversee the issue, revision, updating, and deletion of SOPs, SWO/DWO, and timetable trials for new sections	-	-	-	-
PC45. facilitate the initiation of revenue train services, including pilot train movement, ATO/UTO/ATB mode operations, and AFTC functionality checks	-	-	-	-
PC46. execute procedures for negotiating neutral sections, managing pantograph failures, and monitoring Panto-OHE interface cameras	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC47. oversee train stabling, depot entry/exit, train movement, and fitness checks	-	-	-	-
PC48. conduct emergency evacuation using UG facilities, mid-tunnel shafts, and walkways for detrainment of passengers due to technical failure, explosion, or fire	-	-	-	-
PC49. operate and manage non-ATP vehicles such as RGM and CMV/CTMC and address incidents like rail fractures, weld failures, rollback scenarios, radio communication issues and resetting of passenger alarms in train	-	-	-	-
PC50. coordinate vehicle movement on link lines and rescue immobilized trains	-	-	-	-
PC51. respond to fire/smoke incidents, including UG train fires, ensuring passenger and staff safety	-	-	-	-
PC52. operate ETS/BLS plungers, remove foreign objects like kites from OHE, and follow OHE safety guidelines	-	-	-	-
NOS Total	20	20	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N9901
NOS Name	Implement metro operations training and safety procedures
Sector	Logistics
Sub-Sector	
Occupation	Foundation – Metro (Rail)
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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DMR/LSC/N0201: Oversee the operation and maintenance of metro signalling and train control systems

Description

This unit covers the competencies required to operate, monitor, and maintain metro signalling and train control systems, ensuring safe and efficient train movement and responding to system faults with effective troubleshooting and communication

Scope

The scope covers the following :

- Metro signalling
- Communication-Based Train Control (CBTC) System
- Platform Screen Doors (PSD) and depot operations
- Markers, indicators, overlap, isolation, and sighting distance
- Schedule of Dimension (SoD)
- Control table and interlocking systems
- Overview of interlocking (IXL) systems
- Point machine working, importance and general definition
- Point and crossing
- Signal Overview (Semaphore to CLS)
- Track detection devices
- Automatic Train Supervision (ATS)
- Automatic Train Protection (ATP)
- Working of relays
- Maintenance precautions
- Telecom and signalling interfacing
- P-Way, traction, and E&M integration
- Rolling stock and signalling system interface
- Visits and practical exposure

Elements and Performance Criteria

Metro signalling

To be competent, the user/individual on the job must be able to:

- PC1.** define key signalling terms and use them accurately in communication with team members
- PC2.** identify the functioning of S&T systems like signal, point, track circuit, axle counter, interlocking, ESP, EKT, ATP, and ATS

Communication-Based Train Control (CBTC) System

To be competent, the user/individual on the job must be able to:

- PC3.** operate CBTC systems to control train movement, ensuring adherence to safety protocols
- PC4.** use CBTC to monitor train location, position, and movement through communication channels

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- PC5.** demonstrate the working of the CBTC moving block concept and how it manages train spacing to the relevant personnel
- PC6.** troubleshoot basic issues in the CBTC system and escalate unresolved issues to relevant technical support

Platform Screen Doors (PSD) and depot operations

To be competent, the user/individual on the job must be able to:

- PC7.** inspect platform screen doors to ensure proper alignment and functionality
- PC8.** respond to any PSD malfunctions and initiate emergency protocols, as required
- PC9.** facilitate depot signalling operations, managing train movements for maintenance activities
- PC10.** execute test track signalling operations, testing dynamic train functions such as ATC and braking

Markers, indicators, overlap, isolation, and sighting distance

To be competent, the user/individual on the job must be able to:

- PC11.** identify and interpret different types of marker boards and indicators used in metro signalling to control and monitor train movements
- PC12.** demonstrate the function of markers and indicators to the relevant personnel for guiding train operations, including start, stop, and caution signals
- PC13.** implement procedures for overlap and isolation to ensure safe distance and section control
- PC14.** measure sighting distances to ensure train operators have adequate visibility in all operating conditions

Schedule of Dimension (SoD)

To be competent, the user/individual on the job must be able to:

- PC15.** refer to the SoD to ensure signalling installations comply with safety and clearance requirements
- PC16.** apply SoD standards in practical scenarios, adjusting installation procedures to maintain regulatory compliance

Control table and locking arrangements

To be competent, the user/individual on the job must be able to:

- PC17.** use control tables to monitor and configure signal and track locking arrangements
- PC18.** identify different types of locking arrangements using the control table

Overview of interlocking (IXL) systems

To be competent, the user/individual on the job must be able to:

- PC19.** differentiate between various interlocking systems, e.g. mechanical, Route Relay Interlocking (RRI), Solid State Interlocking (SSI), and Computer-Based Interlocking (CBI)
- PC20.** explain the basic working principles, operational differences, advantages and limitations of different IXL systems

Point machine working, importance and general definition

To be competent, the user/individual on the job must be able to:

- PC21.** explain the working principles of trailable and non-trailable point machines
- PC22.** conduct checks on point machines to ensure correct alignment and smooth switching

Signal Overview (Semaphore to CLS)

To be competent, the user/individual on the job must be able to:

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PC23. differentiate between Semaphore and Colour Light Signals (CLS) signalling based on the type and purpose

PC24. operate and interpret various types of signals used in metro operations, ensuring compliance with modern CLS protocols

Track detection devices

To be competent, the user/individual on the job must be able to:

PC25. operate track circuits and axle counters, monitoring for accurate train detection and section occupancy

PC26. differentiate between track circuit and axle counter functionalities, explaining their operational advantages

PC27. conduct routine maintenance, calibration and troubleshooting on track circuits and axle counters to ensure reliability

Automatic Train Supervision (ATS)

To be competent, the user/individual on the job must be able to:

PC28. operate ATS commands to control train movements, respond to alarms, and manage schedules

PC29. monitor ATS status, ensuring all local and central ATS protocols are active and accurate

Automatic Train Protection (ATP)

To be competent, the user/individual on the job must be able to:

PC30. ensure correct activation of ATP operations during train movements

PC31. monitor ATP status and report any deviations from safety standards immediately

Working of relays

To be competent, the user/individual on the job must be able to:

PC32. inspect relay connections and settings to verify they meet operational standards for controlling signal and track circuit functions

PC33. perform basic troubleshooting on relays to diagnose common issues, such as contact wear or coil failure, and report findings

Maintenance precautions

To be competent, the user/individual on the job must be able to:

PC34. conduct general maintenance checks on signalling equipment following safety protocols

PC35. document all maintenance activities, including preventive and corrective tasks, in the logbook

PC36. coordinate with other departments and report on the status of signalling maintenance

Telecom and signalling interfacing

To be competent, the user/individual on the job must be able to:

PC37. ensure telecom systems are functioning correctly and troubleshoot connectivity with signalling systems

PC38. conduct tests on the fiber optic transmission system to ensure robust network communication

PC39. maintain and troubleshoot networking connections between telecom and signalling systems

PC40. utilize systems available for emergency communication effectively

P-Way, traction, and E&M integration

To be competent, the user/individual on the job must be able to:

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- PC41.** inspect P-Way interfaces with signalling systems to ensure accurate alignment and track quality
- PC42.** verify traction system components (OHE, RSS, TSS) and manage safety requirements during power transitions
- PC43.** oversee traction power requirements and interfaces, ensuring coordinated responses with signalling operations
- PC44.** carry out OHE operation during maintenance and failures
- PC45.** carry out lift rescue operation for passengers trapped in lift
- PC46.** start and stop escalators during following operational procedures during Standard scenarios/emergencies

Rolling stock and signalling system interface

To be competent, the user/individual on the job must be able to:

- PC47.** check the signalling and rolling stock interfaces for continuous data exchange
- PC48.** monitor CCTV integration with signalling for safe train operations and incident management

Visits and practical exposure

To be competent, the user/individual on the job must be able to:

- PC49.** participate in hands-on training by visiting signalling equipment rooms, gaining practical exposure to equipment functions and maintenance protocols
- PC50.** conduct supervised visits to the Depot Control Centre and Central ATS to observe train control operations in a real-time environment
- PC51.** visit the Fault Management Controller (FMC) and central ATS for a comprehensive understanding of signalling operations and incident management

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the hierarchy structure of (S&T) O&M and project wing
- KU2.** the primary functions of the S&T department in metro operations to ensure safe, efficient, and reliable train movement, and to increase the line capacity
- KU3.** the duties of S&TC junior engineers
- KU4.** the role of signalling systems in metro operations, and safety standards they support
- KU5.** the key signalling terms/definitions (e.g. interlocking, track circuits, axle counters) and systems including ATP, ATS, Electronic Token (EKT), and advanced technologies
- KU6.** different S&T systems used in DMRC
- KU7.** the basic concepts of the CBTC system
- KU8.** the operational role of CBTC moving block concept, and troubleshooting of common CBTC issues
- KU9.** the design, alignment, functionality of PSDs, and emergency protocols for passenger safety
- KU10.** the systems of working adopted in metro railway
- KU11.** various modes of train operations
- KU12.** the movement protocols within the depot to ensure safe and efficient transit

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- KU13.** the signaling procedures for directing rakes to maintenance areas and across interlocked zones
- KU14.** the use of the test track for evaluating the performance of the Train Borne ATC System
- KU15.** the applications of specific operational modes within the depot
- KU16.** the roles of various marker boards and indicators in metro signalling, including start, stop, and caution markers
- KU17.** the function of overlap, isolation and sighting distance in signalling
- KU18.** SoD standards and their significance in ensuring safe train operation and signalling compliance
- KU19.** the function and application of control tables in safe, efficient train movements
- KU20.** different types of locking arrangements, i.e. approach locking, signal locking, track locking
- KU21.** the working principles and operational differences in different interlocking systems (i.e. mechanical, RRI, SSI, CBI), including advantages and limitations
- KU22.** the definitions and working principles point machines
- KU23.** the application of trailable and non-trailable point machines in route setting and switch operations
- KU24.** the structure and function of points and crossings in managing train routing and track allocation
- KU25.** the evolution of signalling systems from Semaphore to CLS, and the benefits of CLS in modern metro networks
- KU26.** the basic function of various train detection arrangements used in Delhi Metro, i.e. AC Track Circuit, DC Track Circuit and Axle Counters
- KU27.** the operational basics of track circuits versus axle counters, including detection, reliability, and suitability for various sections of track
- KU28.** the basic functions of local and central ATS system
- KU29.** ATS functionality, Non-CBTC and CBTC ATS simulators, and commands/ response protocols for system alarms
- KU30.** different grades of operations in the ATP
- KU31.** key relay functions, types, and their role in ensuring safe train movements
- KU32.** the preventive and corrective maintenance practices and documentation requirements for signalling systems
- KU33.** telecom systems, emergency communication, and network troubleshooting
- KU34.** basic concept, principle, structure, advantages, disadvantages and applications of Fiber Optics Transmission System (FOTS)
- KU35.** the modes of transmission and troubleshooting of FOTS
- KU36.** the basic concepts of networking, including hardware, software, protocols and connection medium
- KU37.** basic networking terminology, computer networks, types of Computer Network Architecture, network devices, network topology, OSI model, protocol, IP addresses, Mac addresses, DNS server, etc
- KU38.** the networking troubleshooting procedures
- KU39.** the basic concepts of P-Way and its components including speed restriction, rail fracture, points & crossing, and track patrolling with respect to signalling

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- KU40.** P-Way and OHE interfacing with signalling, alignment and track checks, and safety standards for transitions
- KU41.** the purpose of RSS, TSS, ASS, and AMS
- KU42.** the components of OHE
- KU43.** OHE with respect to signalling and its interfacing with signalling system
- KU44.** the do's and don'ts of OHE
- KU45.** SCADA and TVS coordination with train control and signalling
- KU46.** lift and escalator functionality, emergency protocols, and response procedures for station safety
- KU47.** the features and systems of rolling stock
- KU48.** the types of rolling stock used in metros
- KU49.** the advantages of CBTC over Non-CBTC system along with its interfacing with train CCTV
- KU50.** continuous data exchange between signalling and rolling stock, with CCTV monitoring
- KU51.** the layout, standards for maintenance, and roles of Depot Control Centers, central ATS, and FMC in metro operations
- KU52.** interfacing of rolling stock with respect to Non-CBTC and CBTC system along with interfacing of train CCTV
- KU53.** key operational protocols and maintenance practices for equipment within signalling rooms, Depot Control Centre, and Fault Management Controller (FMC)
- KU54.** the importance of practical exposure at facilities like the Depot Control Centre, FMC, and Central ATS to reinforce knowledge and readiness

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** make quick, safe, and informed decisions in dynamic situations, such as signalling faults and train movement adjustments
- GS2.** promptly identify issues in signalling systems, markers, and telecommunications, recognizing potential safety risks or operational inefficiencies
- GS3.** troubleshoot equipment and systems, including CBTC, ATS, and PSD, and determining corrective actions or escalation needs for unresolved issues
- GS4.** document maintenance activities, incident reports, and technical logs with clear and accurate detail
- GS5.** explain technical issues, signalling terms, and system operations clearly to team members and stakeholders, ensuring mutual understanding
- GS6.** prioritize tasks, manage maintenance schedules, and respond to system issues efficiently to minimize downtime and maintain safety
- GS7.** practice attention to detail in monitoring signalling systems, ensuring accurate calibration, and verifying system integrity during inspections
- GS8.** collaborate smoothly with multi-disciplinary teams, including telecom, traction, and P-Way personnel, fostering a cohesive work environment
- GS9.** interpret control tables, interlocking configurations, and train detection data to maintain safe operations

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Not
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Qualification Pack

- GS10.** adjust quickly to operational demands, respond to emergencies, and shift focus based on system requirements and safety protocols

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Metro signalling</i>	1	1	-	0.5
PC1. define key signalling terms and use them accurately in communication with team members	-	-	-	-
PC2. identify the functioning of S&T systems like signal, point, track circuit, axle counter, interlocking, ESP, EKT, ATP, and ATS	-	-	-	-
<i>Communication-Based Train Control (CBTC) System</i>	1	1	-	0.5
PC3. operate CBTC systems to control train movement, ensuring adherence to safety protocols	-	-	-	-
PC4. use CBTC to monitor train location, position, and movement through communication channels	-	-	-	-
PC5. demonstrate the working of the CBTC moving block concept and how it manages train spacing to the relevant personnel	-	-	-	-
PC6. troubleshoot basic issues in the CBTC system and escalate unresolved issues to relevant technical support	-	-	-	-
<i>Platform Screen Doors (PSD) and depot operations</i>	1	1	-	1
PC7. inspect platform screen doors to ensure proper alignment and functionality	-	-	-	-
PC8. respond to any PSD malfunctions and initiate emergency protocols, as required	-	-	-	-
PC9. facilitate depot signalling operations, managing train movements for maintenance activities	-	-	-	-
PC10. execute test track signalling operations, testing dynamic train functions such as ATC and braking	-	-	-	-
<i>Markers, indicators, overlap, isolation, and sighting distance</i>	1	1	-	0.5

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. identify and interpret different types of marker boards and indicators used in metro signalling to control and monitor train movements	-	-	-	-
PC12. demonstrate the function of markers and indicators to the relevant personnel for guiding train operations, including start, stop, and caution signals	-	-	-	-
PC13. implement procedures for overlap and isolation to ensure safe distance and section control	-	-	-	-
PC14. measure sighting distances to ensure train operators have adequate visibility in all operating conditions	-	-	-	-
<i>Schedule of Dimension (SoD)</i>	1	1	-	0.5
PC15. refer to the SoD to ensure signalling installations comply with safety and clearance requirements	-	-	-	-
PC16. apply SoD standards in practical scenarios, adjusting installation procedures to maintain regulatory compliance	-	-	-	-
<i>Control table and locking arrangements</i>	2	2	-	0.5
PC17. use control tables to monitor and configure signal and track locking arrangements	-	-	-	-
PC18. identify different types of locking arrangements using the control table	-	-	-	-
<i>Overview of interlocking (IXL) systems</i>	1	1	-	0.5
PC19. differentiate between various interlocking systems, e.g. mechanical, Route Relay Interlocking (RRI), Solid State Interlocking (SSI), and Computer-Based Interlocking (CBI)	-	-	-	-
PC20. explain the basic working principles, operational differences, advantages and limitations of different IXL systems	-	-	-	-
<i>Point machine working, importance and general definition</i>	2	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. explain the working principles of trailable and non-trailable point machines	-	-	-	-
PC22. conduct checks on point machines to ensure correct alignment and smooth switching	-	-	-	-
<i>Signal Overview (Semaphore to CLS)</i>	1	1	-	0.5
PC23. differentiate between Semaphore and Colour Light Signals (CLS) signalling based on the type and purpose	-	-	-	-
PC24. operate and interpret various types of signals used in metro operations, ensuring compliance with modern CLS protocols	-	-	-	-
<i>Track detection devices</i>	1	1	-	0.5
PC25. operate track circuits and axle counters, monitoring for accurate train detection and section occupancy	-	-	-	-
PC26. differentiate between track circuit and axle counter functionalities, explaining their operational advantages	-	-	-	-
PC27. conduct routine maintenance, calibration and troubleshooting on track circuits and axle counters to ensure reliability	-	-	-	-
<i>Automatic Train Supervision (ATS)</i>	1	1	-	0.5
PC28. operate ATS commands to control train movements, respond to alarms, and manage schedules	-	-	-	-
PC29. monitor ATS status, ensuring all local and central ATS protocols are active and accurate	-	-	-	-
<i>Automatic Train Protection (ATP)</i>	1	1	-	0.5
PC30. ensure correct activation of ATP operations during train movements	-	-	-	-
PC31. monitor ATP status and report any deviations from safety standards immediately	-	-	-	-
<i>Working of relays</i>	1	1	-	0.5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC32. inspect relay connections and settings to verify they meet operational standards for controlling signal and track circuit functions	-	-	-	-
PC33. perform basic troubleshooting on relays to diagnose common issues, such as contact wear or coil failure, and report findings	-	-	-	-
<i>Maintenance precautions</i>	1	1	-	0.5
PC34. conduct general maintenance checks on signalling equipment following safety protocols	-	-	-	-
PC35. document all maintenance activities, including preventive and corrective tasks, in the logbook	-	-	-	-
PC36. coordinate with other departments and report on the status of signalling maintenance	-	-	-	-
<i>Telecom and signalling interfacing</i>	1	1	-	0.5
PC37. ensure telecom systems are functioning correctly and troubleshoot connectivity with signalling systems	-	-	-	-
PC38. conduct tests on the fiber optic transmission system to ensure robust network communication	-	-	-	-
PC39. maintain and troubleshoot networking connections between telecom and signalling systems	-	-	-	-
PC40. utilize systems available for emergency communication effectively	-	-	-	-
<i>P-Way, traction, and E&M integration</i>	1	1	-	0.5
PC41. inspect P-Way interfaces with signalling systems to ensure accurate alignment and track quality	-	-	-	-
PC42. verify traction system components (OHE, RSS, TSS) and manage safety requirements during power transitions	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC43. oversee traction power requirements and interfaces, ensuring coordinated responses with signalling operations	-	-	-	-
PC44. carry out OHE operation during maintenance and failures	-	-	-	-
PC45. carry out lift rescue operation for passengers trapped in lift	-	-	-	-
PC46. start and stop escalators during following operational procedures during Standard scenarios/emergencies	-	-	-	-
<i>Rolling stock and signalling system interface</i>	1	1	-	0.5
PC47. check the signalling and rolling stock interfaces for continuous data exchange	-	-	-	-
PC48. monitor CCTV integration with signalling for safe train operations and incident management	-	-	-	-
<i>Visits and practical exposure</i>	1	1	-	0.5
PC49. participate in hands-on training by visiting signalling equipment rooms, gaining practical exposure to equipment functions and maintenance protocols	-	-	-	-
PC50. conduct supervised visits to the Depot Control Centre and Central ATS to observe train control operations in a real-time environment	-	-	-	-
PC51. visit the Fault Management Controller (FMC) and central ATS for a comprehensive understanding of signalling operations and incident management	-	-	-	-
NOS Total	20	20	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0201
NOS Name	Oversee the operation and maintenance of metro signalling and train control systems
Sector	Logistics
Sub-Sector	
Occupation	Signalling & Telecom, Signalling & Telecom
NSQF Level	4.5
Credits	6
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

DMR/LSC/N0202: Oversee the operation and maintenance of signalling and safety systems for rail operations

Description

This unit covers the operation, maintenance, and troubleshooting of various signalling, safety, and communication systems essential to rail operations. It includes using circuit diagrams, conducting preventive and corrective maintenance on systems such as interlocking, platform screen doors (PSDs), axle counters, point machines, and Automatic Train Protection (ATP), and managing communication systems for safe and efficient train movement

Scope

The scope covers the following :

- Signalling system and circuit drawing
- Measuring tools and PSD
- Interlocking (IXL) systems
- Signals and signal maintenance
- Point machines
- ATS and track access
- ATP equipment
- Power supply and CCTV system
- Radio communication and route table

Elements and Performance Criteria

Signalling system and circuit drawing

To be competent, the user/individual on the job must be able to:

- PC1.** explain the overall signalling system and identify subsystems relevant to the train operations
- PC2.** read and interpret circuit diagrams to trace faults and conduct root-cause analyses
- PC3.** document and report findings from circuit analyses and recommend corrective actions

Measuring tools and PSD

To be competent, the user/individual on the job must be able to:

- PC4.** identify and use measuring tools to confirm signalling equipment parameters, such as voltage and resistance
- PC5.** perform operational checks on PSDs, confirming alignment with signalling control
- PC6.** execute scheduled maintenance on PSDs, documenting any deviations and corrective actions

Interlocking (IXL) systems

To be competent, the user/individual on the job must be able to:

- PC7.** operate interlocking systems, ensuring that all safety interlocks are engaged correctly
- PC8.** maintain interlocking equipment by performing regular checks on locking components and circuits
- PC9.** troubleshoot and resolve basic interlocking faults, escalating complex issues to senior technicians

Qualification Pack

Signals and signal maintenance

To be competent, the user/individual on the job must be able to:

- PC10.** inspect signal visibility, lighting, and positioning, ensuring compliance with visibility standards
- PC11.** conduct routine signal maintenance to ensure each signal is fully operational
- PC12.** respond promptly to signal failures and coordinate with control centers to rectify issues

Axle counter and track circuit

To be competent, the user/individual on the job must be able to:

- PC13.** test axle counter systems for accuracy, ensuring they correctly detect and register train movements
- PC14.** calibrate and maintain track circuits, checking for consistency in section occupancy detection
- PC15.** log and report any anomalies in track detection for further investigation

Point machines

To be competent, the user/individual on the job must be able to:

- PC16.** inspect and test trailable and non-trailable point machines for alignment and proper locking
- PC17.** oversee maintenance of point machines, including scheduled checks and fault rectifications
- PC18.** operate Electric Key Transmission (EKT) and Emergency Stop Plunger (ESP) for safe train movement control
- PC19.** reset EKT, ESP, and SPK (Personnel Protection Key) during emergencies following safety protocols

ATS and track access

To be competent, the user/individual on the job must be able to:

- PC20.** manage ATS commands for normal and critical scenarios, including track access and block granting
- PC21.** coordinate with the Fault Management Controller (FMC) for track access permissions and communicate block statuses

ATP equipment

To be competent, the user/individual on the job must be able to:

- PC22.** monitor onboard & wayside ATP equipment, verifying that it consistently safeguards train operations
- PC23.** use diagnostic and monitoring tools for ATP systems, identifying and resolving basic faults
- PC24.** download and analyze logs from ATP system for preventive maintenance and troubleshooting

Power supply

To be competent, the user/individual on the job must be able to:

- PC25.** test power supply continuity and ensure all electrical safety standards are met, including earthing arrangement
- PC26.** oversee power supply maintenance activities

CCTV system

To be competent, the user/individual on the job must be able to:

- PC27.** check and ensure proper maintenance of CCTV systems, ensuring camera angles and video quality are optimal

Qualification Pack

PC28. identify and ensure prompt replacement of faulty CCTV components to maintain system integrity

Radio communication

To be competent, the user/individual on the job must be able to:

PC29. operate and ensure proper maintenance of radio communication systems for their effective function for operational needs

Route Table

To be competent, the user/individual on the job must be able to:

PC30. check the alignment of route tables with current track configurations, confirming proper signalling setup

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the signalling system's structure, components, and its subsystems critical to train operations
- KU2.** how to read and interpret circuit diagrams, including symbols, layouts, and circuit components, for tracing faults and conducting root-cause analyses
- KU3.** the use of measuring tools, such as multimeters, for testing signalling equipment parameters (e.g., voltage, resistance)
- KU4.** PSD functions, alignment, and integration with signalling controls to ensure safe passenger movement
- KU5.** maintenance tasks for PSDs, including regular checks, troubleshooting, alignment verification, and documentation of deviations
- KU6.** the interlocking system's purpose, features, hardware and software configurations, and safety interlocks
- KU7.** the operational outline of interlocking
- KU8.** interlocking maintenance activities, such as checks on locking components, circuits, and safety mechanisms for bi-weekly, monthly, and quarterly schedules
- KU9.** the signal visibility, lighting, and positioning standards
- KU10.** signal maintenance, including regular inspections, cleaning, adjustment, and prompt response to signal failures
- KU11.** the working principles of axle counters and track circuits, their role in train detection, and fault detection
- KU12.** the maintenance activities for axle counters and track circuits, including calibration, consistency checks, and anomaly reporting
- KU13.** the general definitions, working principles and importance of trailable and non-trailable point machines
- KU14.** the indoor/outdoor functionality, and maintenance schedules for trailable and non-trailable point machines
- KU15.** the basic functions, principles, and operations of EKT, ESP, and SPK for controlling train movement and emergency scenarios
- KU16.** the procedure for resetting EKT, ESP, and SPK during emergencies and the applicable safety protocols

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- KU17.** the ATS systems, including functions with/without UTO/DTO functionality at central and local level, and their operational commands
- KU18.** track access procedures, including block granting and FMC coordination for safe access permissions
- KU19.** the ATP systems, including the role of onboard and wayside equipment
- KU20.** the diagnostic and monitoring tools for ATP troubleshooting, including log downloads, fault detection, and preventive maintenance planning
- KU21.** the power supply setups, including continuity and earthing requirements for signalling system reliability
- KU22.** the CCTV systems, including video quality checks, alignment, and maintenance activities to ensure security and operational readiness
- KU23.** the importance, operational requirements, and maintenance of radio communication systems for optimal functionality
- KU24.** route table alignment with track configurations to support effective train routing and signalling setup

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** make timely and informed decisions regarding fault identification, maintenance actions, and safety protocols, minimizing operational disruptions
- GS2.** identify technical issues across signalling, interlocking, and ATP systems, analyzing root causes, and implement appropriate solutions
- GS3.** document fault findings, maintenance activities, and corrective measures accurately, supporting a robust record-keeping and reporting process
- GS4.** coordinate with team members and communicate track access status with control centers
- GS5.** write clear, concise reports, maintenance logs, and status updates on equipment conditions, ensuring all records are comprehensible for future reference
- GS6.** prioritize and complete tasks within scheduled timeframes, especially during time-sensitive situations like emergency responses or scheduled maintenance windows
- GS7.** analyze circuit diagrams, assess equipment parameters, and evaluate system responses, aiding in fault diagnosis and preventive planning
- GS8.** conduct equipment checks, align components, and perform calibrations, ensuring system safety and functionality at all times
- GS9.** adapt to evolving rail technology to enhance operational efficiency, learning new technical tools and system updates
- GS10.** implement safety protocols, emergency response procedures, and safe work practices, crucial for personal and operational safety during tasks

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Signalling system and circuit drawing</i>	3	1	-	1
PC1. explain the overall signalling system and identify subsystems relevant to the train operations	-	-	-	-
PC2. read and interpret circuit diagrams to trace faults and conduct root-cause analyses	-	-	-	-
PC3. document and report findings from circuit analyses and recommend corrective actions	-	-	-	-
<i>Measuring tools and PSD</i>	2	2	-	1
PC4. identify and use measuring tools to confirm signalling equipment parameters, such as voltage and resistance	-	-	-	-
PC5. perform operational checks on PSDs, confirming alignment with signalling control	-	-	-	-
PC6. execute scheduled maintenance on PSDs, documenting any deviations and corrective actions	-	-	-	-
<i>Interlocking (IXL) systems</i>	2	2	-	1
PC7. operate interlocking systems, ensuring that all safety interlocks are engaged correctly	-	-	-	-
PC8. maintain interlocking equipment by performing regular checks on locking components and circuits	-	-	-	-
PC9. troubleshoot and resolve basic interlocking faults, escalating complex issues to senior technicians	-	-	-	-
<i>Signals and signal maintenance</i>	2	2	-	1
PC10. inspect signal visibility, lighting, and positioning, ensuring compliance with visibility standards	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. conduct routine signal maintenance to ensure each signal is fully operational	-	-	-	-
PC12. respond promptly to signal failures and coordinate with control centers to rectify issues	-	-	-	-
<i>Axle counter and track circuit</i>	1	2	-	1
PC13. test axle counter systems for accuracy, ensuring they correctly detect and register train movements	-	-	-	-
PC14. calibrate and maintain track circuits, checking for consistency in section occupancy detection	-	-	-	-
PC15. log and report any anomalies in track detection for further investigation	-	-	-	-
<i>Point machines</i>	3	3	-	1
PC16. inspect and test trailable and non-trailable point machines for alignment and proper locking	-	-	-	-
PC17. oversee maintenance of point machines, including scheduled checks and fault rectifications	-	-	-	-
PC18. operate Electric Key Transmission (EKT) and Emergency Stop Plunger (ESP) for safe train movement control	-	-	-	-
PC19. reset EKT, ESP, and SPK (Personnel Protection Key) during emergencies following safety protocols	-	-	-	-
<i>ATS and track access</i>	1	2	-	1
PC20. manage ATS commands for normal and critical scenarios, including track access and block granting	-	-	-	-
PC21. coordinate with the Fault Management Controller (FMC) for track access permissions and communicate block statuses	-	-	-	-
<i>ATP equipment</i>	2	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. monitor onboard & wayside ATP equipment, verifying that it consistently safeguards train operations	-	-	-	-
PC23. use diagnostic and monitoring tools for ATP systems, identifying and resolving basic faults	-	-	-	-
PC24. download and analyze logs from ATP system for preventive maintenance and troubleshooting	-	-	-	-
<i>Power supply</i>	1	1	-	0.5
PC25. test power supply continuity and ensure all electrical safety standards are met, including earthing arrangement	-	-	-	-
PC26. oversee power supply maintenance activities	-	-	-	-
<i>CCTV system</i>	1	1	-	0.5
PC27. check and ensure proper maintenance of CCTV systems, ensuring camera angles and video quality are optimal	-	-	-	-
PC28. identify and ensure prompt replacement of faulty CCTV components to maintain system integrity	-	-	-	-
<i>Radio communication</i>	1	1	-	0.5
PC29. operate and ensure proper maintenance of radio communication systems for their effective function for operational needs	-	-	-	-
<i>Route Table</i>	1	1	-	0.5
PC30. check the alignment of route tables with current track configurations, confirming proper signalling setup	-	-	-	-
NOS Total	20	20	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0202
NOS Name	Oversee the operation and maintenance of signalling and safety systems for rail operations
Sector	Logistics
Sub-Sector	
Occupation	Signalling & Telecom, Signalling & Telecom
NSQF Level	4.5
Credits	7
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on a knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training centre (as per the assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practicals for every student at each examination/training centre based on this criterion.
6. To pass the Qualification Pack, every trainee should score a minimum of 60% of aggregate marks to successfully clear the assessment.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 60

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DMR/LSC/N9901.Implement metro operations training and safety procedures	20	20	0	10	50	25
DMR/LSC/N0201.Oversee the operation and maintenance of metro signalling and train control systems	20	20	-	10	50	25
DMR/LSC/N0202.Oversee the operation and maintenance of signalling and safety systems for rail operations	20	20	-	10	50	25
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	25
Total	80	90	0	30	200	100

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Not
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Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.